Blockchain Data Availability: Benchmarking Throughput, Economics, Utilization, and Security of Six Production Protocols

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Abstract

Data availability (DA) bandwidth bounds the throughput of rollups and sharded blockchains. This paper presents a benchmarking study of six production DA protocols (Polkadot ELVES, Ethereum EIP-4844/PeerDAS, Celestia, Espresso Tiramisu, NEAR, and Avail) using 92 days of mainnet data (November 2025 to February 2026). We introduce a taxonomy that classifies DA architectures into three paradigms: data availability sampling (DAS), verifiable information dispersal (VID), and validator-set recovery. Under a unified methodology we measure throughput (MiB/s), cost ($/MiB), utilization, and block dynamics across all six protocols. Key findings are: (i) window-mean utilization of the protocol maximum ranges from 0.18% (NEAR) to 35% (Ethereum); Ethereum ran at ∼60% of its maximum (∼90% of its fee-market target) until the BPO forks raised capacity, and the other five protocols stayed below 3%; (ii) median DA cost spans nearly seven orders of magnitude, from $\$3.1\times10^{-9}$/MiB (Espresso, a fixed placeholder fee) to $0.020/MiB (Celestia), although the six cost measures differ in what they include; (iii) no protocol slashes for pure data withholding, so enforcement is indirect in every design; and (iv) no protocol guarantees permanent availability: default retention ranges from ∼25 hours (Polkadot) to ∼18 days (Ethereum), while Avail and Espresso nodes retain data unless operators enable pruning. Ethereum's Fusaka upgrade and two blob-parameter-only forks fall inside the window, allowing before–after comparisons of fee-market behaviour. We release all collection tooling, datasets, and analysis code under open licenses.

Keywords: data availability, blockchain scalability, erasure coding, data availability sampling, benchmarking, rollups, distributed systems, measurement

1. Introduction

A blockchain can scale only if every participant can be confident that block data has been published and can be retrieved. This property, *data availability* (DA), is a security requirement for rollups, parachains and sharded designs. Without it, validators cannot verify state transitions, light clients cannot detect fraud, and the system's trust model reduces to that of a permissioned committee [1].

Several DA designs have reached production in the past three years. Ethereum introduced blob-carrying transactions (EIP-4844) and, with the Fusaka upgrade in December 2025, activated PeerDAS for column sampling by full nodes [2, 3]. Celestia operates a dedicated DA network with two-dimensional erasure coding and namespace-filtered sampling [4]. Polkadot's ELVES protocol distributes erasure-coded parachain data across its validator set with approval-gated finality [5]. Espresso's Tiramisu introduces a three-layer Verifiable Information Dispersal (VID) architecture that uses algebraic dispersal proofs instead of DAS [6]. NEAR provides DA through Nightshade sharding, on which rollups post data as function-call arguments [7], and Avail combines KZG polynomial commitments with DAS-enabled light clients on a Substrate-based chain [8].

Published comparisons of these designs are qualitative surveys [1, 9], project documentation, or measurements of a single protocol [10, 11]. We are not aware of a study that measures throughput, cost and utilization across several DA networks with one methodology; no common empirical baseline therefore exists for comparing DA backends.

This paper makes the following contributions:

1. A taxonomy that classifies DA protocols by verification paradigm (data availability sampling, verifiable information dispersal, or validator-set recovery), encoding scheme and commitment type (Section 2).
2. A comparison of encoding redundancy, commitment schemes (KZG, NMT and Merkle) and verification mechanisms across the six protocols, based on their specifications and documentation (Section 3).
3. Measurements from 92 days of mainnet data (November 2025–February 2026) for Polkadot, Ethereum, Celestia, Espresso, NEAR and Avail, covering throughput (MiB/s), cost ($/MiB), utilization and block dynamics under common definitions (Section 5).
4. A comparison of trust assumptions, Byzantine fault tolerance thresholds, withholding enforcement and retention policies (Section 6).
5. The collection scripts, raw datasets and analysis code, released under Apache 2.0 (code) and CC BY 4.0 (data).

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2. Background and Related Work

2.1. Data Availability Fundamentals

Data availability (DA) must be distinguished from the related but distinct problems of data *storage* and data *retrieval*. Storage concerns the persistence of data over time; retrieval concerns the ability to read stored data on demand. DA concerns whether the data was published at all. A block producer who withholds part of a block's data can cause validators and light clients to accept an invalid state transition, because the withheld data may contain a fraudulent transaction whose invalidity can never be proven if the data is unavailable [12].

Definition 1 (Data Availability)**.** *A block B is available if, within a bounded time Δ after its header is published, the complete block data can be reconstructed from the pieces held by honest participants. A DA scheme lets a participant who does not download B decide whether B is available, with bounded error (cf. the completeness and soundness notions of [13]).*

A DA scheme must provide four properties:

1. **Completeness.** All data constituting block B is published to the network. A DA scheme is complete if no honest node accepts a block header unless the corresponding data has been made available to the network.
2. **Soundness.** The encoding of B is verifiably correct. Given a commitment $C(B)$ included in the block header, any participant can verify that a received coded piece is consistent with $C(B)$ without possessing the entire block.
3. **Availability detection.** Resource-constrained participants (light clients) can assess whether the data behind a block header is available without downloading the full block. This property separates DA from simple data retrieval.
4. **Reconstruction.** The original data B can be recovered from any sufficiently large subset of coded pieces. If $E(B)$ denotes the erasure-coded encoding of B producing n coded pieces with redundancy factor r, then for a maximum-distance-separable (one-dimensional) code B is reconstructible from any n/r pieces. For Celestia's two-dimensional code, recovery in the worst case requires more than $(2k)^2 - (k+1)^2$ of the $(2k)^2$ shares ($\sim$75%).

Notation. Throughout this paper we use the following conventions. A block's raw data is denoted B, with $|B|$ its size in bytes. The encoding function $E(B)$ produces n coded pieces (variously called shares, chunks, or cells depending on the protocol). A commitment $C(B)$ is included in the block header and serves as a binding digest of $E(B)$. Verification $V(C, s_i)$ checks that coded piece s_i is consistent with commitment C. The redundancy factor $r = n \cdot |s_i|/|B|$ is the inverse of the code rate; it excludes replication of pieces across nodes.

2.2. Taxonomy of DA Approaches

We classify the six protocols along three dimensions: the *availability verification paradigm* (how clients gain confidence that data is available), the *encoding scheme* (how raw data is expanded into coded pieces: 1D Reed–Solomon [14], 2D Reed–Solomon, or polynomial VID), and the *commitment scheme* (how pieces are bound to the header: KZG [15], Namespaced Merkle Trees, or plain Merkle trees). Section 3 compares encodings and commitments in detail; here we define the three verification paradigms.

DAS (Avail, Celestia, Ethereum/PeerDAS). Light clients request a few random coded pieces; if all arrive with valid proofs, the client concludes with high probability that the full data is available. Security rests on an *honest-minority* assumption: at least one honest, non-eclipsed full node must seed correctly encoded pieces into the network.

Validator-set recovery (Polkadot, NEAR). The validator set collectively stores erasure-coded chunks and attests to availability through consensus (Polkadot's availability bitfields [5], NEAR's per-shard chunk producers [7]); there is no client-side sampling.

VID (Espresso). Verifiable information dispersal [16] combines erasure coding with algebraic proofs that each committee member received a valid piece; a threshold of signatures forms a DA certificate [6]. Light-client confidence derives from the certificate, not from sampling.

2.3. Protocols Under Study

We summarize each protocol as deployed during our window (November 2025–February 2026); Table 3 gives the parameters.

Polkadot (ELVES). The relay chain erasure-codes parachain Proof-of-Validity (PoV) data across ~600 validators [5, 17]; a block is available once two-thirds sign availability bitfields, after which it enters approval checking and GRANDPA finality. Up to 100 cores each carry a PoV of up to 10 MiB every 6 s. DA is priced through coretime rather than per byte, and chunks are kept for ~25 hours.

Ethereum (EIP-4844 and PeerDAS). Blobs of 128 KiB are committed with KZG and priced by a separate exponential blob-gas market [2]. Pectra (EIP-7691 [18]) set a target/maximum of 6/9 blobs per block. The Fusaka upgrade on 3 December 2025 activated PeerDAS (EIP-7594 [3]), with 2× column extension over 128 columns, and EIP-7918 [19], under which excess blob gas is not reduced while the blob base fee is below BLOB_BASE_COST × base_fee_per_gas/GAS_PER_BLOB, which stops the fee from falling further (a reserve price). Two blob-parameter-only (BPO) forks then raised the target/maximum to 10/15 (9 December) and 14/21 (7 January 2026). Blobs are pruned after 4,096 epochs (~18 days).

Celestia. A $k \times k$ original data square is extended to $2k \times 2k$ and committed with Namespaced Merkle Trees, enabling namespace-filtered retrieval [4]. CometBFT gives single-slot finality every ~6 s. App version 6 activated on 24 November 2025, and on 3 December the maximum square rose from 128×128 (8 MiB) to 256×256 (32 MiB). Light nodes sample within a 7-day window (CIP-036).

Espresso (Tiramisu). Tiramisu layers a KZG-based VID scheme (Savoiardi), a DA committee for fast retrieval (Mascarpone), and an untrusted CDN for bulk delivery [6]; it is now marketed as EspressoDA. HotShot gives responsive finality of ∼2 s and the maximum block size was 1 MB throughout our window. Mainnet 0 runs 100 nodes operated by 20 parties without slashing. Retention is set per node: pruning is off by default and, when enabled, targets 7 days.

NEAR. Nightshade shards erasure-code chunk data among each shard's chunk producers [7]. Rollups post data as function-call input to a blob-store contract; it is pruned after the 5-epoch garbage-collection window (∼1.5 days at the ∼0.6 s block time in force since May 2025). A single transaction is capped at 1.5 MiB. There is no DAS and no active withholding challenge.

Avail. Row-wise 1D Reed–Solomon encoding with KZG commitments supports DAS by light clients over a DHT [8]. BABE/GRANDPA produce a block every 20 s; application-specific data retrieval lets clients download only their application's data. The maximum block size was 4 MiB throughout the window. Nodes prune *state* after 256 blocks but keep block data by default, so there is no protocol-defined DA expiry; the VectorX bridge relays DA attestations to Ethereum with a ∼2-hour delay.

2.4. Related Work

Foundational theory. Croman et al. [20] quantified the throughput and latency limits of monolithic blockchains, motivating the separation of data publication from execution. Al-Bassam et al. [12] introduced fraud and data availability proofs, the foundation for DAS: by erasure-coding block data and letting light clients randomly sample coded pieces, a network can guarantee availability with high probability under an honest-minority assumption. Hall-Andersen, Simkin, and Wagner formalized the security definition of DAS [13] and proposed FRIDA, a DAS construction without a trusted setup [21]. Coded Merkle Trees [22] and the ACeD oracle [23] reduce the cost of detecting incorrect coding. On the dispersal side, asynchronous verifiable information dispersal [16] is the primitive behind committee-based designs, and Nazirkhanova, Neu, and Tse [24] adapted it to rollups using KZG commitments [15]; Fisch et al. [25] extend VID to permissionless settings.

Protocol-specific designs. Al-Bassam's LazyLedger proposal [4] introduced a dedicated DA blockchain with namespace-filtered retrieval, which evolved into Celestia. Burdges et al. [5] formalized Polkadot's ELVES protocol for availability and validity, proving security under Byzantine assumptions with approval-based checking. Bearer et al. [6] describe the HotShot consensus and Tiramisu DA architecture. Ethereum has progressed from blob transactions (EIP-4844 [2]) through blob throughput increases (EIP-7691 [18]) to peer-level DAS (EIP-7594 [3]), an incremental path toward Danksharding [26]; Chaudhuri et al. [27] and Pigaglio et al. [28] study the networking design space of Ethereum DAS through simulation.

Table 1: Encoding and redundancy.

Protocol	Encoding	Redund.	Reconstruction
Polkadot [5]	1D RS (validators)	∼3×	f+1 of n (n=3f+k)
Ethereum [3]	1D RS (PeerDAS cells)	∼2×	50%+ of 128 columns
Celestia [4]	2D RS ($k \times k \to 2k \times 2k$)	∼4×	>75% of shares (worst case)
Espresso [6]	Polynomial VID	$1/r$, r=m/n	m of n evaluations
NEAR [7]	1D RS (chunk parts)	∼3×	1/3 of parts
Avail [8]	1D RS (row-wise)	∼2×	n of 2n coded pieces

Empirical and economic studies. Several recent studies measure Ethereum's blob market in isolation. Crapis et al. [29] model the economics of EIP-4844; Heimbach et al. [10] and Park et al. [30] analyse blob fee dynamics and the impact of EIP-4844; and Lee et al. [11] measure blob utilisation over 180 days. All of these focus on Ethereum alone.

Surveys. Saif et al. [1] and Thibault et al. [9] survey DA and rollup designs, classifying approaches by trust assumptions and encoding strategies, without measurements from production networks.

Positioning of this work. To our knowledge, no prior study measures six production DA protocols under one methodology. Prior empirical work covers Ethereum alone, and prior comparative work is qualitative or simulation-based. Our study complements the theoretical results above and the surveys of [1, 9] with a cross-protocol empirical baseline that includes Ethereum's DA market across the Fusaka (PeerDAS) and BPO transitions.

3. Protocol Architecture Comparison

We compare the six protocols on encoding and redundancy, commitment scheme, and verification paradigm. These choices fix a protocol's bandwidth cost, what a light client can check on its own, and which parties must be trusted.

3.1. Encoding and Redundancy

All six protocols use Reed–Solomon (RS) erasure coding, but they differ in dimensionality, redundancy factor and how data is reconstructed. Table 1 summarizes these differences.

Higher redundancy costs proportionally more bandwidth and aggregate storage; in return, data can be recovered when a larger fraction of coded pieces is missing.

Celestia has the highest redundancy at ∼4× due to its two-dimensional RS structure: original data of dimension $k \times k$ is extended to $2k \times 2k$, yielding four times the original data volume in shares [4]. The two-dimensional layout keeps bad-encoding fraud proofs at $O(\sqrt{n})$ size, since a proof needs only one row or column. The 2D layout also allows a single row or column of length $2k$ to be recovered from any k of its shares, so a node can repair one row without reconstructing the whole block.

Polkadot and NEAR both use ∼3× redundancy, distributing erasure-coded pieces across their respective validator sets. In Polkadot's ELVES protocol, parachain block data (Proofs of Validity, or PoVs) are RS-encoded and distributed such that

Table 2: Commitment scheme properties (KZG: Avail, Ethereum, Espresso; NMT: Celestia; Merkle: Polkadot, NEAR).

Property	KZG	NMT	Merkle
Proof size	$O(1)$, ~48 bytes	$O(\log n)$	$O(\log n)$
Trusted setup	Yes (SRS)	No	No
Namespace filter	No	Native	No
Verification	Pairings	Hashes	Hashes
Encoding correctness	Algebraic	Fraud proofs	Not verifiable

each validator stores one coded chunk, with reconstruction requiring $f+1$ of n chunks where $n = 3f + k$ [5, 17]. NEAR's Nightshade sharding similarly encodes chunk data with ~3× redundancy, requiring only 1/3 of the parts for reconstruction [7]. In both, responsibility for the data is spread over the validator set; there is no separate committee and no sampling.

Ethereum and Avail have the lowest redundancy, ~2×. Under PeerDAS, each blob's 4,096 field elements are extended to 8,192 evaluations and grouped into 128 cells of 2,048 bytes, one per column; any 64 columns suffice for reconstruction [3]. Avail performs row-wise RS encoding on its application data matrix, extending n original pieces to $2n$ coded pieces per row, so that any n of the $2n$ pieces suffice for recovery [8]. With a 2× extension any half of the coded pieces recovers the data, which together with KZG commitments is enough for DAS.

Espresso's Tiramisu uses a polynomial-based verifiable information dispersal (VID) scheme. Data is represented as a polynomial that is evaluated at n distinct points (one per committee member); any m evaluations suffice for interpolation, where the code rate $r = m/n$ is configurable [6]. Algebraically this is an RS code, but because it is built into the VID commitment layer, one proof certifies both that the encoding is correct and that each member received its piece.

3.2. Commitment Schemes and Proof Systems

The commitment scheme determines what a light client can verify, whether a trusted setup is needed, and the cost of producing and checking proofs. The six protocols use three families: KZG polynomial commitments, Namespaced Merkle Trees (NMTs) and standard Merkle trees. Table 2 compares their properties.

KZG polynomial commitments [15] are used by Avail, Ethereum, and Espresso. A KZG commitment to a polynomial $p(x)$ of degree d is a single elliptic curve group element (~48 bytes on BLS12-381), and an opening proof for any evaluation $p(z) = v$ is also a single group element. Verification requires two pairing operations, which are computationally more expensive than hash evaluations but yield $O(1)$-size proofs regardless of the data size. KZG commitments enable *algebraic verification of encoding correctness*: given commitments to the rows (or columns) of an erasure-coded data matrix, anyone can verify that the extended portions are consistent with the original data by checking polynomial relationships between commitments, without accessing the underlying data. This lets Avail and Ethereum light clients reject incorrectly encoded data without fraud proofs. The drawback is that KZG needs a structured reference string (SRS) from a trusted setup; anyone holding the setup's secret could forge proofs. Ethereum and Avail both ran multi-party ceremonies to reduce this risk, Ethereum's with over 140,000 contributors.

Celestia uses Namespaced Merkle Trees (NMTs) [4]. An NMT is a standard Merkle tree augmented with namespace identifiers: each leaf is tagged with a namespace, internal nodes carry the minimum and maximum namespace of their subtrees, and the tree is sorted by namespace. This structure enables *namespace inclusion and exclusion proofs*. A light client can request all shares belonging to a specific namespace (e.g., a particular rollup's data) and receive a Merkle proof that the returned set is complete (no shares from that namespace have been omitted, and no extraneous shares have been included). Rollup nodes use this to download only their own data instead of the whole block. NMT proofs are $O(\log n)$ in size. Encoding correctness, however, is *not* algebraically verifiable from the NMT alone. Instead, Celestia relies on Bad Encoding Fraud Proofs (BEFPs) to detect and propagate evidence of incorrect erasure coding [12].

Polkadot and NEAR use standard Merkle trees. Polkadot commits to erasure-coded PoV chunks via a Merkle trie whose root is included in the relay chain block header [17, 31]. NEAR similarly uses Merkle roots to commit to chunk data within shard blocks. Merkle inclusion proofs are $O(\log n)$ and need only hashing to verify. Merkle proofs support sampling (Celestia's NMT is a Merkle tree) but cannot attest to encoding correctness without fraud proofs: a proof shows that a piece belongs to the committed set, not that the erasure coding was done correctly. Polkadot and NEAR do not deploy sampling. In both Polkadot and NEAR, encoding correctness is enforced through the validator set (validators who reconstruct and re-encode can detect errors), but this verification is not accessible to light clients or external observers.

3.3. Verification Paradigms: DAS, VID, and Validator Recovery

The three paradigms below differ in their trust assumptions and in what a light client can check.

3.3.1. Data Availability Sampling (DAS)

DAS, as formalized by Al-Bassam et al. [12], enables light clients to verify data availability without downloading the full block. A light client randomly requests s coded pieces from the network; if all s are returned, the client gains probabilistic confidence that the full data is available. If a fraction q of the pieces is withheld, s independent uniform samples detect the withholding with probability $1 - (1 - q)^s$.

Its 2D RS structure means that light clients sample from the $2k \times 2k$ extended square. Because an adversary must withhold at least ~25% of the extended square to prevent reconstruction, each sample detects withholding with probability $\geq 1/4$, so 16 random samples give a Celestia light client approximately $1 - (3/4)^{16} \approx 99\%$ confidence that the data is available, assuming that at least one honest full node is non-eclipsed and participates in the peer-to-peer network [4].

Avail implements DAS over its 1D RS-encoded rows with KZG commitments. Avail light clients sample random cells from the data matrix; each cell is accompanied by a KZG opening proof. With 10 random samples, an Avail light client reaches a confidence of $1 - 2^{-10} \approx 99.9\%$ [8]. This figure bounds the fraction of cells that are available; it does not bound recoverability under targeted withholding, because the extension is one-dimensional: withholding just over half of the extended cells of a single line makes that line unrecoverable, while a uniform sample hits a withheld cell with probability of only about $1/(2L)$ for a matrix of L such lines. Because each cell is checked against the committed row polynomial, the client also verifies encoding correctness without fraud proofs. Avail light clients fetch cells from a Kademlia DHT run by light clients and fall back to full-node RPC; cells fetched over RPC are written back into the DHT, so later requests can be served by other light clients.

Ethereum activated DAS through PeerDAS (EIP-7594) in the Fusaka upgrade on 3 December 2025. Under PeerDAS, each extended blob is split into 128 columns, and each node custodies a subset of columns determined by its node ID. In the Fulu specification every node custodies at least 4 of 128 custody groups (`CUSTODY_REQUIREMENT`), nodes with attached validators custody at least 8 plus one per 32 ETH of balance (`VALIDATOR_CUSTODY_REQUIREMENT`), and every node samples 8 columns per slot (`SAMPLES_PER_SLOT`) [3]. Each node samples only a few columns, so availability of every column depends on custody being spread across many honest nodes. Any node holding at least 64 columns can reconstruct the remaining columns and re-seed them to the network; before PeerDAS, every consensus node downloaded all blob data [3]. PeerDAS sampling is performed by full nodes; Ethereum has no light-client DAS.

DAS in Celestia and Avail, and column sampling in Ethereum, rely on two assumptions: (i) at least one honest, non-eclipsed full node must hold the complete data and respond to sampling requests, and (ii) sufficient sampling participation must exist in the network to ensure that withholding is detected before the data propagation window closes. An adversary capable of eclipsing a light client's peer connections can cause the client to falsely conclude that data is available (or unavailable) regardless of the true state.

3.3.2. Validator-Set Recovery

In Polkadot and NEAR the validator set stores and serves the erasure-coded data, and availability is attested through consensus instead of sampling.

Polkadot's ELVES protocol ties availability to finality [5]. After a parachain collator produces a block candidate, the relay chain validators erasure-code the Proof of Validity (PoV) and distribute one chunk per validator. Each validator that successfully receives and stores its chunk signs an *availability bitfield* (a bitmap indicating which PoV chunks it holds). A PoV is considered available when $\geq 2/3$ of validators have signed its bitfield. A backed candidate is included only after $\geq 2/3$ of the availability bitfields attest to chunk custody; candidates that miss the availability timeout are dropped. GRANDPA finality is then gated on approval checking of the included candidates [5, 17]. Under the $2/3$ honest-majority assumption, an included candidate is recoverable from the chunks held by honest validators.

NEAR's Nightshade sharding distributes chunk data across shard validators (chunk producers). Each chunk producer is responsible for producing and serving the chunk parts for its assigned shard. A block includes a chunk only once its parts have been distributed to the block producers that track the shard; enforcement relies on this inclusion rule rather than on sampling. The Nightshade design also specifies challenges for invalid or unavailable chunks [7]; we did not measure whether challenges occur on mainnet.

Because availability evidence is a precondition for inclusion (Polkadot) or block production (NEAR), validators have a direct incentive to store and serve data. Light clients, however, cannot check availability themselves: a Polkadot or NEAR light client must trust that the $\geq 2/3$ supermajority attested honestly, with only the validators' stake behind that trust.

3.3.3. Verifiable Information Dispersal (VID)

Espresso's Tiramisu replaces DAS with algebraic dispersal proofs [6]. It has three layers:

- **Savoiardi (VID layer):** Data is encoded as a polynomial and evaluated at n points (one per committee member). A vector commitment (using KZG) binds the evaluations to the polynomial commitment, so each member can verify that it received a correct evaluation. Because shares are dispersed across the whole stake-weighted validator set, an adversary must corrupt a large fraction of total stake, rather than a small committee, to make certified data unrecoverable.
- **Mascarpone (DA committee):** A small committee stores the full block and provides fast retrieval; VID shareholders who receive and verify their evaluations sign, and a DA certificate is formed from a threshold of these signatures.
- **Cocoa (CDN layer):** A secondary retrieval path through an untrusted CDN that provides availability even if committee members go offline, with cryptographic proofs linking the fallback data to the original VID commitments.

There is no sampling: a block is finalized only with a DA certificate. VID guarantees that once a certificate forms, the data is recoverable from any sufficiently large set of honest shareholders; it does not prevent a signer from deleting its share after signing.

Retrieval, however, requires cooperation from the committee or the Cocoa CDN. There is no mechanism for an independent light client to sample data and verify availability without relying on the committee's attestations. Espresso argues that DAS adds "unnecessary redundancies" given VID's algebraic guarantees [6]; as a result, retrievability depends on the committee and the CDN.

3.3.4. No Slashing for Data Withholding

None of the six protocols slashes for pure data withholding. Each has some way to detect or discourage it, but enforcement is always indirect:

- **Polkadot:** Validators are slashed for casting invalid *validity* votes (attesting that an invalid parachain block is valid), but there is no slashing for failing to store or serve erasure-coded chunks. A validator that withholds its chunk does not set its bit in the availability bitfield, which can delay inclusion but does not make an included candidate unrecoverable [5].
- **Celestia:** Bad Encoding Fraud Proofs (BEFPs) allow any full node to prove that the block producer performed incorrect erasure coding [12]. However, BEFPs address encoding correctness, not data withholding. A block producer that correctly encodes data but subsequently withholds it faces no direct slashing penalty.
- **Ethereum:** PeerDAS enforces availability through peer scoring and disconnection: validators that fail to serve their custody columns receive negative gossip scores and may be disconnected from the p2p network. There is no protocol-level slashing for custody failures [3].
- **Espresso:** In the current Mainnet 0 deployment, enforcement is reputational only. No slashing mechanism is implemented for committee members that fail to serve their VID evaluations [6].
- **NEAR:** The Nightshade design specifies challenges for unavailable chunks [7], but no slashing for withholding; enforcement relies on the chunk-inclusion rule.
- **Avail:** Slashing is implemented for equivocation (double-signing) but not for data withholding. Validators that fail to make sampled cells available face no direct economic penalty [8].

Proving that a node *has* data but *refuses* to serve it is harder than proving that it served *incorrect* data. Withholding is an omission fault, and attributing an omission to malice instead of a network partition, latency or node failure is an open problem in distributed systems.

Table 3 collects these parameters. Celestia and Avail let light clients verify availability themselves, and Ethereum's PeerDAS samples at full nodes; all three depend on network-level assumptions; validator-recovery (Polkadot, NEAR) and VID (Espresso) designs give deterministic guarantees that hold only if the validators or committee are honest. The commitment scheme follows from the paradigm: KZG gives Ethereum, Avail and Espresso algebraic encoding checks, Celestia pairs sampling with NMTs and fraud proofs, and plain Merkle trees suffice where validators check the encoding themselves. All six use the classical $f < n/3$ Byzantine threshold but apply it differently: Polkadot gates candidate inclusion and finality on it, Ethereum spreads custody, Celestia relies on sampling participation, Espresso on committee certification, NEAR on per-shard honesty, and Avail on sampling backed by algebraic commitments.

4. Methodology

4.1. Data Collection Framework

We collect per-block data from six DA protocols over a 92-day observation window (November 15, 2025 through February 14, 2026) from mainnet RPC endpoints, indexers and public datasets. Coverage is complete except for NEAR, which lacks February 11–14 (88 days, 12.3M of 12.6M blocks), and Espresso and Avail, which lack November 15. Celestia has no TIA price for November 15, so that day is excluded from its cost statistics. All protocol parameters (capacities, fork boundaries, gas schedules, retention defaults) are the values in force at the height of each recorded block.

Table 4 summarizes the data sources, block counts, and collection methods for each protocol.

For each protocol, the collector captures the following:

Polkadot. We scan relay chain blocks via the Polkadot.js RPC API, extracting three event types per block: `paraInclusion.CandidateBacked`, `paraInclusion.CandidateIncluded`, and `paraInclusion.CandidateTimedOut`. We also parse the `paraInherent.enter` extrinsic to extract signed availability bitfield counts and per-core availability fractions. The relay chain does not expose PoV sizes (only commitments), so throughput is estimated as included $\times$ $\overline{\text{PoV}}$/cadence, with $\overline{\text{PoV}}$ the inclusion-weighted parachain-side PoV estimate (Section 4.3); included $\times$ maxPovBytes is kept as an upper bound. Cost data is collected separately from the Coretime chain, scanning `broker.Purchased`, `broker.Renewed`, and `broker.SaleInitialized` events, as well as relay chain `onDemand.OnDemandOrderPlaced` events.

Ethereum. We collect per-block header fields (`blob_gas_used`, `excess_blob_gas`) from the public BigQuery table `bigquery-public-data.crypto_ethereum.blocks` across the full 92-day window, which spans four regimes: Pectra (target 6, max 9), Fusaka (same blob parameters, plus PeerDAS and the EIP-7918 reserve price, from block 23,935,694 on December 3, 2025), BPO1 (target 10, max 15, from block 23,975,778 on December 9), and BPO2 (target 14, max 21, from block 24,179,383 on January 7, 2026). Era boundaries are the first blocks at or after each fork's activation slot. The blob base fee is computed from the header's `excess_blob_gas` using the EIP-4844 exponential pricing formula with the update fraction of the active era. ETH/USD prices are sampled hourly from CoinGecko.

Celestia. Block data is retrieved from the Celenium indexer API, which provides 22 columns per block including `blobs_size`, `blobs_count`, `square_size`, `fill_rate`, and `fee_utia`. The window spans app version 5 and version 6 (from 24 November 2025), and a maximum-square increase from 128 to 256 (8 to 32 MiB) on 3 December 2025; we compute capacity with the maximum in force on each day.

Table 3: Feature and parameter comparison of the six DA protocols during the observation window.

Feature	Polkadot	Ethereum	Celestia	Espresso	NEAR	Avail
Encoding	1D RS (validators)	1D RS (PeerDAS)	2D RS ($k \to 2k$)	Poly. VID	1D RS (chunks)	1D RS (rows)
Commitment	Merkle trie	KZG	NMT	KZG + vector	Merkle	KZG
Redundancy	~3×	~2×	~4×	Configurable	~3×	~2×
DAS support	No	Yes (PeerDAS)	Yes (2D)	No (VID)	No	Yes (1D)
Light client DAS	No	No (full-node sampling)	Yes (16 samples)	No	No	Yes (10 samples)
Block time	6 s (relay)	12 s	6 s	~2 s	~0.6 s	20 s
Max DA capacity	10 MiB×100	2.63 MiB (BPO2)	8→32 MiB*	1 MB	4 MiB×9	4 MiB
Protocol max MiB/s	~167	0.09→0.22	1.33→5.31*	~0.48	~59	0.20
Finality	12–60 s	~12.8 min	~6 s	~2 s	~2 s	20–60 s
DA retention	25 h	~18 d	7 d (sampling)	Node-set	~1.5 d	Node-set†
Fee model	Weight-based	Blob market	Gas-based	Per-byte	Gas-based	Weight-based
Withholding	Inclusion gate	Peer scoring	BEFPs (encoding)	Reputational	Inclusion rule	None
Byzantine threshold	$f < n/3$	$f < n/3$	$f < n/3$	$f < n/3$	$f < n/3$/shard	$f < n/3$

*Celestia's maximum square rose from 128 to 256 (8 to 32 MiB) on 3 Dec 2025. †Avail prunes state after 256 blocks, but block data is kept by default (Substrate `archive-canonical` block pruning).

Table 4: Data sources and block counts (92-day window).

Protocol	Source	Blocks	Key Fields	Price Source
Polkadot	Relay chain RPC	~1.3M	candidates backed/included, bitfields, core index	CoinGecko (DOT)
Ethereum	Google BigQuery	~658K	blob_gas_used, excess_blob_gas, blob count	CoinGecko (ETH)
Celestia	Celenium API	~1.3M	blobs_size, square_size, fill_rate, fee_utia	CoinGecko (TIA)
Espresso	Query-service API	~2.9M	size_bytes, num_transactions, block_time_ms	CoinGecko (ETH)
NEAR	NEAR Lake S3 + RPC	~12.3M	total_encoded_bytes, gas_used, gas_price	CoinGecko (NEAR)
Avail	Substrate RPC	~389K	submit_data_bytes, block_fee_plancks	CoinGecko (AVAIL)

Espresso. We fetch all blocks in the window from the Espresso query-service explorer API. Each block record contains `size_bytes`, `num_transactions`, and `block_time_ms`. HotShot is a responsive consensus protocol, so block times vary. The base fee is 1 wei/byte in every header we sampled between heights 1K and 10.3M.

NEAR. Blocks and chunk headers are read from NEAR Lake (S3), with NEAR RPC as a fallback. Per-block fields include `num_shards`, `total_encoded_bytes` (sum of chunk encoded lengths), `total_gas_used`, and `gas_price`.

Avail. The collector uses the Polkadot.js API (Avail is Substrate-based) to extract `submit_data_bytes`, `submit_data_count`, and `block_fee_plancks` (from `TransactionFeePaid` events) per block. Runtime version (`spec_version`) is tracked to account for parameter changes.

4.2. Metric Definitions

We apply four groups of metrics to every protocol.

4.2.1. Throughput (MiB/s)

We distinguish three throughput measures:

- **Protocol maximum**: the theoretical ceiling given current parameters. Protocol-specific formulas:

$$T_{\max}^{\text{Polkadot}} = \frac{\text{effective_cores} \times \text{max_pov}}{\text{cadence}} \tag{1}$$

$$T_{\max}^{\text{Ethereum}} = \frac{\text{max_blobs} \times 128\ \text{KiB}}{\text{slot_time}} \tag{2}$$

$$T_{\max}^{\text{Celestia}} = \frac{\text{max_data_bytes}}{\text{avg_block_time}} \tag{3}$$

  where effective_cores = $\min(\text{num_cores}, \lfloor \text{validators}/5 \rfloor)$ and cadence is 6s (async backing) for Polkadot.

- **Observed throughput**: measured from production data as

$$T_{\text{obs}} = \frac{\sum \text{payload_bytes}}{\Delta t \times 2^{20}} \quad \text{(MiB/s)} \tag{4}$$

  aggregated at hourly and daily granularity.

- **Utilization**: $U = T_{\text{obs}}/T_{\max}$, representing the fraction of available DA capacity in use.

4.2.2. Cost ($/MiB)

The protocols price DA in different units, which we normalize as follows:

- **Per-byte fee markets** (Celestia, Espresso, Avail): cost is computed from on-chain fees and token prices. For Celestia the daily quantiles use the gas price × 8 gas/byte and exclude the fixed PayForBlobs gas; VWAP divides all fees by blob bytes. For Avail, cost is the transaction fee of each

`submitData` call divided by its payload, so small submissions dominate the per-MiB figure.

$$\text{cost_per_MiB} = \text{fee_per_byte} \times 2^{20} \times \text{token_USD} \quad (5)$$

- **Blob gas market** (Ethereum): cost is the blob base fee from EIP-4844 exponential pricing; it excludes the execution and priority gas of the carrying transaction:

$$\text{cost_per_MiB} = \frac{\text{blobBaseFee}_{\text{gwei}} \times 2^{20}}{10^9} \times \text{ETH_USD} \quad (6)$$

- **Coretime pricing** (Polkadot): DA is bundled with execution capacity. We compute:

$$\text{cost_per_MiB} = \frac{\text{coretime_price_DOT} \times \text{DOT_USD}}{\text{max_MiB_per_region}} \quad (7)$$

  for bulk coretime, and fee_paid/max_pov_MiB for on-demand. This is a capacity price: it assumes a full PoV in every block.

- **Function-call gas** (NEAR): NEAR has no blob type or DA fee market; rollups post data as function-call arguments. We price DA as the marginal gas of a cross-account call carrying 1 MiB of input, from the in-window gas schedule (identical at protocol versions 81 and 82): per byte, the not-same-receiver send cost plus execution plus the cost of reading the input (`env::input`); plus fixed receipt and call costs. That is ~60 Tgas per MiB, multiplied by each block's gas price. State storage (1 NEAR per 100 KB) is avoided because input is pruned. NEAR's chain-wide average fee per byte is reported separately and is not a DA cost.

We report spot cost at percentiles (p50, p90, p99) and volume-weighted average price (VWAP) over the 92-day window.

4.2.3. Annualized Cost

Where a protocol drops DA data after a fixed window, keeping the data available means reposting it periodically. We define:

$$\text{cost_per_MiB_year} = \text{cost_per_MiB} \times \frac{365}{\text{retention_days}} \quad (8)$$

We define retention uniformly as the default horizon after which a non-archival node no longer serves the data: Polkadot 25 h (availability store), NEAR ~1.5 d (5-epoch garbage collection), Ethereum ~18 d (blob sidecars), and Celestia 7 d (the light-node sampling window; bridge and full nodes store 30 days by default). Avail retains block bodies by default and Espresso does not prune unless configured, so neither has a default horizon and we report their spot cost only. The resulting figures are illustrative repost costs, not prices offered by the protocols.

4.2.4. Block Dynamics

We measure block time distributions (p10, p50, p90), block size distributions, and empty block rates. For protocols with variable block times (Espresso's responsive HotShot consensus), we report the empirical distribution rather than a fixed value.

4.3. Normalization and Comparability

Comparing these protocols raises six methodological issues:

1. **Demand-constrained measurements**: window-mean utilization ranges from 0.18% to 35%; only Ethereum exceeded 3%, at ~60% of its maximum before BPO1. Observed throughput therefore reflects demand, and we report protocol maximum throughput alongside it.
2. **Token price volatility**: USD-denominated costs are sensitive to token prices, which fluctuated during our window (daily means: ETH \$1,821–\$3,357; TIA \$0.31–\$0.82). We use contemporaneous hourly prices from CoinGecko rather than fixed spot rates, and report VWAP alongside percentile distributions.
3. **Pricing units**: Polkadot sells blockspace (coretime), Ethereum prices blob gas, and Celestia and Espresso charge per byte. We convert all to \$/MiB. For Polkadot two biases act in opposite directions: amortizing over maximum PoV capacity understates the cost per MiB actually used (~34× at 2.9% utilization), while attributing the whole coretime price to DA overstates the DA share. The figure is a capacity price, not an observed DA cost.
4. **Estimated throughput for Polkadot**: the relay chain does not expose PoV sizes (only commitments). We estimate the mean PoV from parachain-side proof size and extrinsic bytes for 17 parachains covering 47% of inclusions, weighted by inclusion share, and multiply by included candidates. The proxy is uncompressed, so it overstates the real PoV; Section 7.6 gives the range.
5. **Protocol upgrades within window**: Ethereum underwent three upgrades (Fusaka, BPO1, BPO2) and Celestia transitioned eras during our observation period. We segment analysis by protocol era where applicable and compare the periods before and after each transition (Section 5.4).
6. **Throughput measurement conventions**: Two definitional choices affect comparability with other sources (e.g., L2Beat). First, for Ethereum we report protocol maximum throughput using the *absolute maximum* blob count per block (21 under BPO2), whereas some trackers use the EIP-4844 *target* blob count (14 under BPO2), below which fees trend toward zero. The target is the fee-market equilibrium and the maximum is the hard protocol ceiling. Second, for Celestia we measure *user data throughput* (the original data square, ODS), not the full 2D Reed–Solomon encoded square. The extended square is 4× larger (a $k \times k$ ODS becomes $2k \times 2k$ after encoding), so sources that report total encoded bandwidth will show approximately 4× our figures. We report user data throughput because it reflects the amount of application-level data that can be posted per unit time.

Table 5: 92-day throughput and utilization summary.

Protocol	Max MiB/s	p50 MiB/s	Mean MiB/s	Mean util.
Polkadot	167	4.84	4.85	2.9%
Ethereum	0.094→0.22	0.051	0.051	35%
Celestia	1.33→5.31	0.016	0.08	1.7%
Espresso	0.48	0.0012	0.0049	1%
NEAR	59	0.11	0.11	0.18%
Avail	0.2	0.0014	0.0016	0.81%

Polkadot throughput uses the higher of two parachain-side PoV estimates; the lower gives 1.2 MiB/s (Section 7.6). Ethereum and Celestia maxima and utilization use the parameters in force on each day; Max shows the range where the maximum changed.

Table 6: Cost summary over the window ($/MiB). Quantiles are over daily medians.

Protocol	p50	p90	p99	VWAP	Measure
Polkadot	$5.3\times10^{-6*}$	2.7×10^{-5}	3.4×10^{-5}	—	Capacity price
Ethereum	0.0073	0.023	0.074	0.18	Blob base fee
Celestia	0.02	0.03	0.042	0.025	Gas price
Espresso	3.1×10^{-9}	3.4×10^{-9}	3.5×10^{-9}	3.1×10^{-9}	Fixed fee
NEAR	0.01	0.011	0.015	0.0099	Marginal gas
Avail	0.0096	0.032	1.05	0.0074	Per-tx average

*14 of 92 days had bulk coretime purchases; price amortized over maximum PoV capacity. Because quantiles are over daily medians, p99 is close to the highest daily median. VWAP is total spend divided by total volume. The measures differ in what they include (Section 4.2).

4.4. Reproducibility

Collection and analysis code, the raw per-block data and the figures are released under Apache 2.0 (code) and CC BY 4.0 (data) at `https://github.com/Chainscore/score-da`, archived at `https://doi.org/10.5281/zenodo.23276229`. All tables and figures are regenerated from the raw data by the released scripts.

5. Empirical Results

5.1. Observation Window and Protocol Eras

Our window (November 15, 2025 to February 14, 2026) contains several parameter changes, which allow before–after comparisons:

- **Ethereum**: four regimes: Pectra (target 6/max 9 blobs), Fusaka (same blob parameters, plus PeerDAS and the EIP-7918 reserve price, from Dec 3), BPO1 (target 10/max 15, from Dec 9), and BPO2 (target 14/max 21, from Jan 7).
- **Celestia**: app version 6 from Nov 24, and a maximum-square increase from 128 to 256 (8 to 32 MiB per block) on Dec 3.
- **Polkadot, Espresso, NEAR, Avail**: no major parameter changes.

Table 3 lists the active parameters.

5.2. Throughput and Utilization

Fig. 1(a) overlays 7-day rolling throughput for all six protocols on a logarithmic axis, with dashed lines at each protocol maximum; Fig. 1(b) shows utilization on a shared linear axis. Table 5 gives aggregate statistics.

Ethereum has the highest utilization: ∼60% of its maximum under Pectra and Fusaka (∼90% of the target), falling to ∼31% after BPO1 and ∼23% after BPO2 as capacity grew faster than demand. The other five protocols stay below 3% on average: Polkadot's estimated data utilization is ∼3% (per relay block, ∼26 of 100 cores carry an included candidate; Section 5.5), Celestia's is ≤1% outside the December load test, and Espresso, Avail and NEAR are at or below 1%. Observed throughput therefore reflects demand; capacity is given by the protocol maximum (Table 5). Volumes over the window were 9–37 TiB for Polkadot (estimated), ∼787 GiB for NEAR, ∼614 GiB for Celestia and ∼392 GiB for Ethereum. These figures are not like for like: NEAR's is total chunk data (transactions and receipts) of a general-purpose chain, not rollup DA payload, and Polkadot's includes parachain state witnesses.

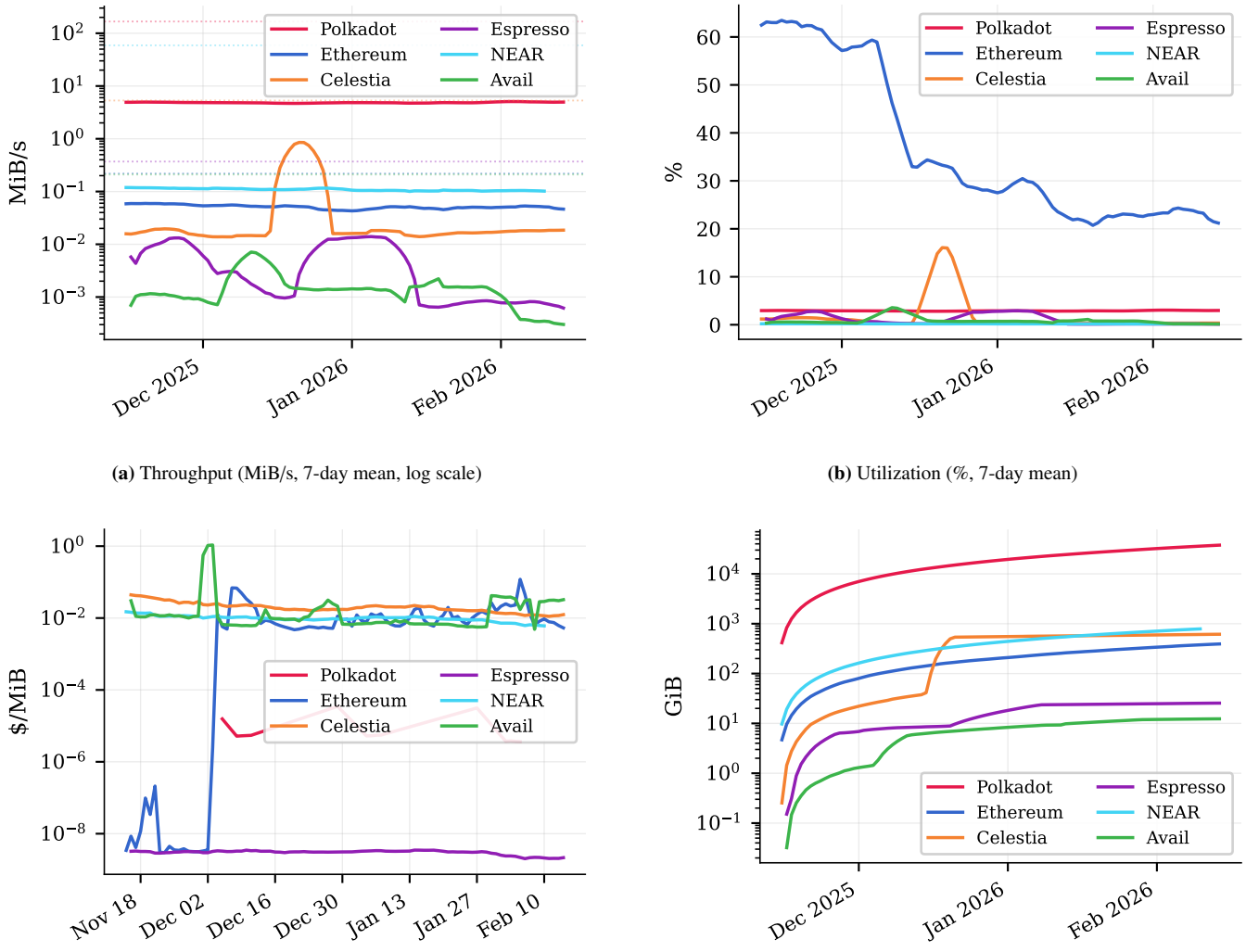
5.3. Cost Analysis

Fig. 2 shows per-protocol daily cost quantile bands (p10/p50/p90) and Table 6 summarizes the distributions.

Median DA cost spans nearly seven orders of magnitude, from $\$3.1\times10^{-9}$/MiB on Espresso to $0.020/MiB on Celestia, reflecting differences in pricing mechanism. Espresso's fee has been fixed at 1 wei/byte since genesis, so its cost follows ETH/USD, falling from $\sim\$3.5\times10^{-9}$ to $\sim\$2\times10^{-9}$ as ETH fell; in the permissioned Mainnet 0 it is a placeholder and does not reflect demand.

Before Fusaka, Ethereum's blob base fee sat at the 1-wei floor on ∼28% of blocks and the median cost was $\sim\$4\times10^{-9}$/MiB. From the first day after Fusaka no block is at the floor and the median rises six orders of magnitude to ∼$0.009/MiB. BPO1 had not yet added capacity, so the change is consistent with EIP-7918, the only fee-relevant change at that boundary (Section 5.4). Ethereum's VWAP ($0.18/MiB) is far above its median because two days, November 18 and December 4, account for 78% of window blob spend; excluding them, VWAP is $0.040/MiB. The December 4 spike (daily p90 $0.56/MiB) is a transient overshoot immediately after Fusaka activation.

Avail's median ($0.0096/MiB) is close to Celestia's, but its daily median exceeds $1/MiB on December 1–3, days of low volume (45–50 MiB/day) and many small submissions; its p90 stays near $1.1/MiB through late December. This is consistent with per-transaction fees dominating the per-MiB cost of small submissions; we do not isolate the contribution of Avail's fee modifier. NEAR has no blob type or DA fee market, so we price DA as the marginal gas of posting 1 MiB as function-call input (Section 4.2): ∼0.006 NEAR, or ∼$0.010/MiB at the window's prices. The gas price stayed at its 10^8 yoctoNEAR minimum throughout, so the USD cost moves only with the NEAR price. NEAR's chain-wide average fee per byte of chunk data is far higher (∼$0.53/MiB) because most of its gas pays for contract execution, not data. Polkadot's bulk coretime price, observed on 14 days, is below $\$10^{-4}$/MiB on a capacity basis; it covers execution as well as DA.



(a) Throughput (MiB/s, 7-day mean, log scale)

(b) Utilization (%, 7-day mean)

(c) Median cost ($/MiB, log scale)

(d) Cumulative volume (GiB, log scale)

Figure 1: Cross-protocol comparison over the window. Five protocols stay below 3% of capacity; Ethereum reaches ~60% before BPO1. Celestia's December 16–21 partner load test peaks at ~22% of its quadrupled capacity. Polkadot throughput and volume are estimates from parachain-side PoV sizes (Section 4.3).

Table 7: Illustrative annualized cost: p50 spot cost × reposts per year at the default retention horizon (Section 4.2).

Protocol	Retention	Reposts/yr	p50 $ /MiB/yr
NEAR	~1.5 d	237	2.37
Celestia	7 d	52	1.02
Ethereum	~18 d	20	0.15
Polkadot	25 h	350	0.0019
Espresso	node-set	—	(spot 3.1×10^{-9})
Avail	node-set	—	(spot 0.0096)

5.3.1. Retention-Adjusted Cost

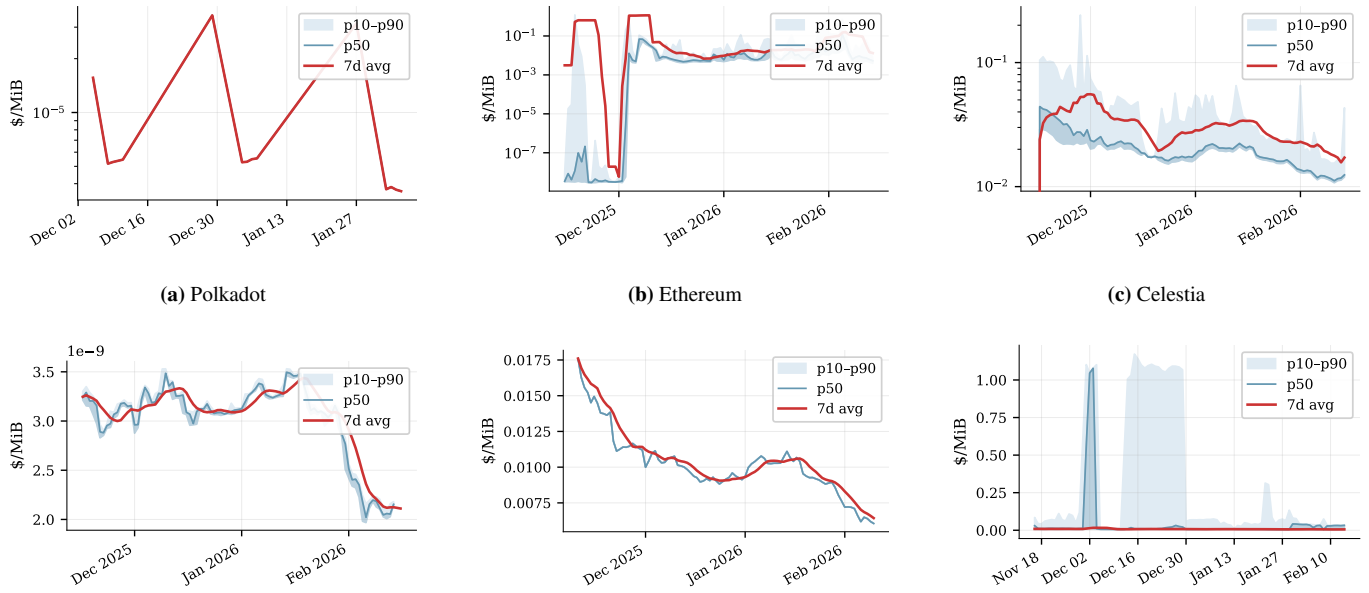
Table 7 and Fig. 3(b) normalize cost by retention (Section 4.2).

Short retention multiplies cost: NEAR's 5-epoch garbage-collection window (~1.5 days) requires ~237 reposts per year and gives the highest annualized cost (~$2.4/MiB/yr, about twice Celestia's), while Celestia's 7-day sampling window lifts its $0.02/MiB spot cost to ~$1/MiB/yr. Ethereum's 18-day window keeps its annualized cost near $0.15/MiB/yr, and Polkadot's cost stays negligible even after ~350 reposts. Avail and Espresso define no protocol-level expiry, so their cost cannot be annualized on the same basis.

5.4. Protocol Upgrades

5.4.1. Ethereum: Pectra → Fusaka → BPO1 → BPO2

Fig. 4 separates the effects of the three upgrades. Under Pectra the blob fee was at the 1-wei minimum on 5–68% of blocks per day. Fusaka's EIP-7918 ties a reserve price to the execution base fee; from December 4 no block is at the floor (Fig. 4(c)) and the median cost rises to ~$0.009/MiB, while utilization



(a) Polkadot **(b)** Ethereum **(c)** Celestia

(d) Espresso **(e)** NEAR **(f)** Avail

Figure 2: Daily cost quantile bands ($/MiB) per protocol. (a) Polkadot bulk coretime, on days with purchases only. (b) Ethereum: fees sit at the 1-wei floor until Fusaka (Dec 3), then settle near $0.005–0.02/MiB. (c) Celestia declines from ~$0.05 to ~$0.012/MiB, partly with the TIA price. (d) Espresso's fixed 1 wei/byte fee tracks ETH/USD. (e) NEAR: marginal cost of posting a blob as function-call input, set by the gas schedule and the minimum gas price. (f) Avail has the widest spread, with p50 above $1/MiB on December 1–3.

stays at ~58%. BPO1 and BPO2 each lowered utilization (from ~58% to ~31% to ~23%) and target utilization (from ~87% to ~47% to ~34%; Fig. 4(b)). The median cost stayed between $0.007 and $0.010/MiB in both BPO eras (Fig. 4(a)), consistent with fees held near the EIP-7918 reserve price rather than set by congestion; we did not collect the execution base fee, so this is not verified against the reserve price directly. Daily blob volume stayed at ~4.1–4.7 GiB in all four eras, so the added capacity was not used during the window.

5.4.2. Celestia: December Load Test

Celestia raised its maximum square from 128 to 256 on December 3, quadrupling capacity to ~5.3 MiB/s. From December 16 to 21, daily throughput rose eighty-fold, from a baseline of ~0.015 MiB/s to ~1.2 MiB/s (~7.3 MiB per block). Celestia's status page announced on December 16 that a partner would run mainnet tests adding about 1 MB/s; the load began a few hours before the notice. It reached ~22% of the new ceiling, and would have used ~90% of the pre-December capacity of 1.33 MiB/s. Cost did not rise: the daily median fell slightly ($0.020 to $0.017/MiB) and the daily p90 fell from ~$0.046 to ~$0.018/MiB, close to the median, as the test posted large blobs at the minimum gas price. Afterwards throughput and the p90 returned to their earlier levels; the largest load in the window did not produce congestion.

5.5. Block Dynamics and Polkadot Core Occupancy

Fig. 5(a–b) shows block time distributions for the two protocols where they are informative; the fixed-cadence protocols (Ethereum 12 s, Polkadot 6 s, Avail 20 s) are omitted. Celestia holds 5–6 s after an elevated p90 in mid-November. Espresso's responsive HotShot consensus has p10 ~1 s and p50 ~2 s, with a p90 that falls from ~6.5 s in the first weeks to ~3 s by February.

Fig. 5(c) shows Polkadot's per-block core occupancy: ~26 of 100 cores carry an included candidate in a typical relay block. Per-block occupancy understates the number of cores in use: on-chain core descriptors assign ~58–62 cores to bulk coretime (50–55 parachains) and the remaining ~40 to the on-demand pool, and on the sampled days (November 18, January 6, February 10) 41 distinct cores carried at least one candidate. Many parachains produce a block every 12 s and so occupy their core in alternate relay blocks; the unused capacity is mainly the on-demand pool and ~20 bulk-assigned cores with no inclusions.

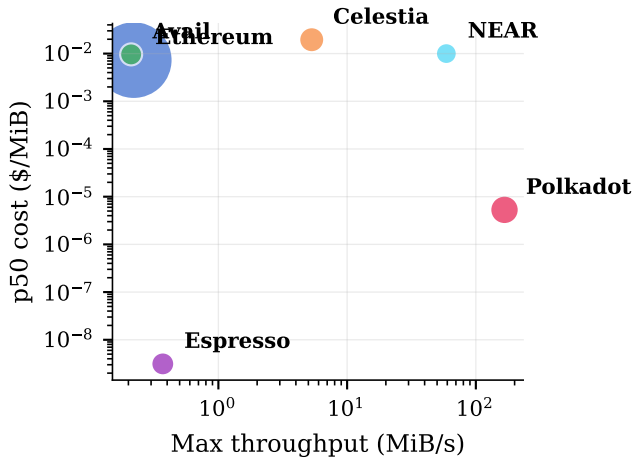
Espresso's payload shows bursts in late November and in late December–early January, then ~40–50 MiB/day; we did not identify the source of the bursts.

6. Security and Trust Analysis

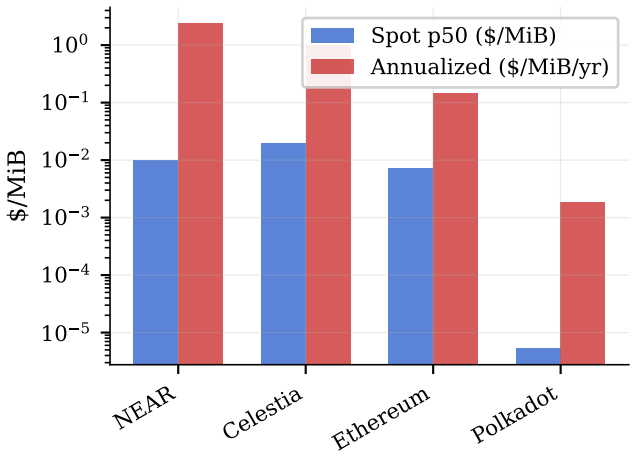
6.1. Byzantine Fault Tolerance

All six protocols inherit the classical BFT bound of $f < n/3$, where f is the number of Byzantine-faulty nodes and n is the total committee or validator set size. How the threshold is applied differs between protocols.

Polkadot. Polkadot's relay chain requires availability bitfields from $\geq 2/3$ of validators before a parachain candidate is included; GRANDPA finality is then gated on approval checking. With approximately 600 active validators, the system tolerates

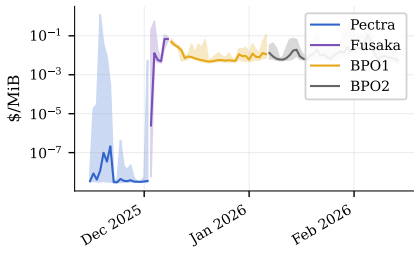


(a) Maximum throughput vs. median cost (bubble size: mean utilization)

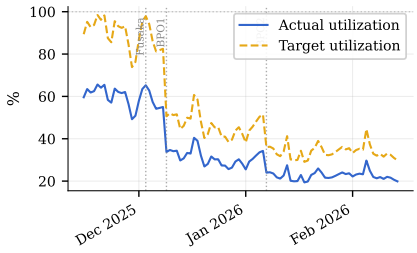


(b) Spot vs. annualized cost

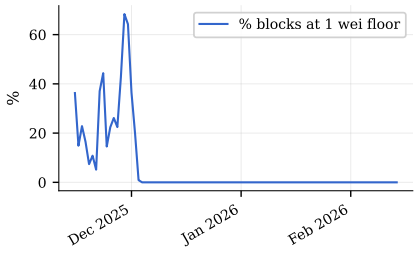
Figure 3: Capacity, cost and retention. Protocols do not lie on a single capacity–cost frontier, and retention reorders the cost ranking.



(a) Cost bands by era ($/MiB)



(b) Target vs. actual utilization (%)



(c) Blocks at the 1-wei fee floor (%)

Figure 4: Ethereum blob market across the Pectra, Fusaka, BPO1 and BPO2 eras. The fee leaves the floor at Fusaka (EIP-7918), before any capacity change; the BPO forks lower utilization without materially changing median cost.

up to ~200 Byzantine faults. ELVES adds approval checking: randomly selected validators re-execute parachain blocks post-inclusion, and a single invalid execution triggers a dispute that can slash the responsible validators [5]. The probability of an invalid block escaping detection decreases exponentially with the number of approval checkers.

Ethereum. Ethereum's Casper FFG requires a 2/3 supermajority of validators (by stake weight) to finalize blocks. The active set had approximately 1 million validators during the window. PeerDAS assigns column custody per node ID, with additional custody groups per 32 ETH of attached validator balance; availability of every column depends on custody being spread across many honest nodes [3].

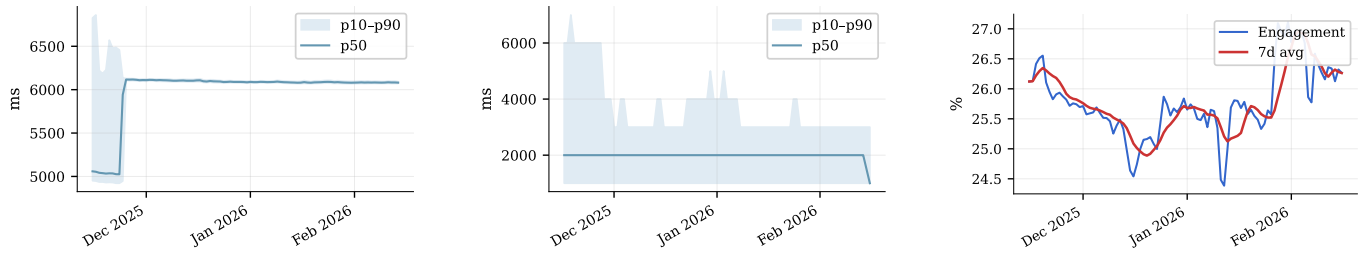
Celestia. Celestia runs CometBFT (Tendermint) consensus, which requires $\geq$ 2/3 of validators (by stake weight) to commit each block. The validator set is capped at 100 in the current configuration. Light-client sampling adds a second check: if validators finalize a block whose data is unavailable, sampling detects the withholding with high probability [12], provided enough light clients take part.

Espresso. Espresso's HotShot consensus operates with 100 nodes across 20 operators in the Mainnet 0 configuration. The DA certificate requires $\geq$ 2/3 of committee signatures, tolerating up to ~33 Byzantine nodes. However, the small operator count (20) means that the effective trust assumption is closer to $f < 7$ at the operator level [6].

NEAR. NEAR's Nightshade design applies the BFT threshold per shard [7]: chunks are produced and validated by the validators assigned to each shard, so the assumption is that fewer than one third of each shard's assigned stake is Byzantine. With 9 shards and approximately 400 validators, the size of each shard's assignment depends on the stake distribution.

Avail. Avail's BABE/GRANDPA consensus requires $\geq$ 2/3 honest validators for both block production and finality. DAS light clients provide independent verification, analogous to Celestia's model. The VectorX bridge to Ethereum adds a further trust dependency: bridge liveness depends on the prover infrastructure, and the approximately 2-hour bridge delay creates a window during which Avail-native finality and Ethereum-bridged attestations may diverge [8].

Table 8 summarizes the BFT characteristics across protocols.



(a) Celestia block time (ms)

(b) Espresso block time (ms)

(c) Polkadot per-block core occupancy (%)

Figure 5: Block time distributions (p10/p50/p90) for Celestia and Espresso, and Polkadot per-block core occupancy.

Table 8: Byzantine Fault Tolerance and Validator Set Characteristics

Protocol	BFT Threshold	Validator Count	Consensus	Finality	Stake
Polkadot	$f < n/3$	~600	BABE + GRANDPA	12–60 s	~1.8M DOT min
Ethereum	$f < n/3$ (stake)	~1M	Casper FFG	~12.8 min	32–2,048 ETH per validator
Celestia	$f < n/3$ (stake)	~100	CometBFT	Single-slot (~6 s)	Variable
Espresso	$f < n/3$	100 (20 operators)	HotShot	~2 s	None (permissioned)
NEAR	$f < n/3$ per shard	~400	Nightshade	~2 s	Seat-price auction
Avail	$f < n/3$	~100	BABE + GRANDPA	20–60 s	Variable

6.2. Data Withholding Enforcement

Section 3.3 showed that no protocol slashes for data withholding; here we compare how each one detects it.

Detection is sampling-based for Celestia, Avail and Ethereum (missing responses to random samples), attestation-based for Polkadot and NEAR (a missing bitfield or chunk endorsement is visible on-chain, but only validator-level withholding is detected), and certificate-based for Espresso (a missing DA certificate blocks finalization).

Enforcement gap. In every protocol, detecting that data is unavailable does not identify who is responsible. A validator that fails to serve a chunk may be offline, network-partitioned, or maliciously withholding, and the protocol cannot distinguish these cases. A plausible reason no protocol slashes for withholding is that slashing on such evidence would also punish honest nodes that have network problems.

6.3. Retention and Persistence

No protocol guarantees permanent availability (Table 3). Four have a default retention horizon on non-archival nodes: Polkadot's availability store keeps finalized chunks for 25 hours, NEAR garbage-collects after 5 epochs (~1.5 days), Celestia light nodes sample within 7 days (CIP-036), and Ethereum serves blobs for 4,096 epochs (~18 days). Avail and Espresso define no network-wide expiry: Avail nodes keep block data by default and Espresso nodes choose their own pruning policy, so availability beyond the latest blocks rests on operators rather than the protocol. An optimistic rollup needs the DA layer to keep data for at least its challenge period, typically 7 days. Polkadot (25 h) and NEAR (~1.5 d) keep it for less and Celestia's 7-day sampling window leaves no margin, so rollups on these layers must archive data themselves or shorten their challenge periods; on Avail and Espresso it depends on node configuration. Bridges that verify DA attestations must also complete within the retention window; Avail's VectorX attestations reach Ethereum about 2 hours after the block. After expiry, data survives only if full nodes, indexers or rollup operators archive it voluntarily.

6.4. Trust Assumptions Summary

Table 9 lists each protocol's trust assumptions.

7. Discussion

7.1. Trade-offs Between the Three Paradigms

Sampling protocols (Celestia, Avail, Ethereum). Celestia and Avail give light clients the strongest guarantees among the six designs, because a resource-constrained node can check availability itself; Ethereum's PeerDAS samples at full nodes only. These guarantees weaken if too few nodes sample, a dependence the other two paradigms do not have.

Validator-recovery protocols (Polkadot, NEAR). Among the six designs, Polkadot's inclusion rule is the most direct enforcement: a candidate is included only after $\geq 2/3$ of availability bitfields attest to chunk custody, and finality is then gated on approval checking. In exchange, light clients cannot verify availability themselves. For parachain validation this is acceptable, because the relay chain validators are the main consumers of the data; an external rollup using Polkadot or NEAR for DA would have to trust the validator set.

VID protocol (Espresso). VID disperses shares across the whole operator set, so making certified data unrecoverable requires corrupting a large fraction of operators rather than a

Table 9: Trust Assumption Comparison Across DA Protocols

Protocol	Availability Assumption	Encoding Trust	Additional Assumptions
Polkadot	$\geq 2/3$ validators honest (bitfield voting)	Validators re-encode and verify (approval checking)	Relay chain liveness
Ethereum	$\geq 2/3$ validators honest (Casper FFG) + sufficient custody distribution	KZG algebraic verification (no fraud proofs needed)	Trusted setup integrity (KZG ceremony)
Celestia	$\geq 2/3$ validators honest + sufficient DAS participation	BEFPs from ≥ 1 honest full node	≥ 1 non-eclipsed honest full node
Espresso	$\geq 2/3$ committee signs DA certificate	KZG + vector commitment algebraic verification	Committee honesty; CDN liveness
NEAR	$< 1/3$ Byzantine per shard	Chunk producers re-encode	Honest shard assignment; no stake concentration
Avail	$\geq 2/3$ validators honest + DAS participation	KZG algebraic verification	Trusted setup integrity; VectorX bridge liveness

small committee. In Mainnet 0 the practical value of this property is limited while the operator set is permissioned (20 operators) and there is no slashing. The 1 wei/byte fee is likewise a placeholder. Whether VID's advantages matter in practice will depend on a permissionless deployment with economic security behind it.

7.2. The Cost–Throughput Frontier

Fig. 3(a) plots median cost against protocol maximum throughput; across the six protocols, cost and maximum throughput are not inversely related. Espresso combines low cost with moderate throughput, but its cost is a Mainnet 0 placeholder. Celestia and Avail have moderate throughput at moderate cost. Polkadot has the highest maximum throughput (~167 MiB/s) at a capacity price of $\sim\$5\times10^{-6}$/MiB, which also covers execution. NEAR combines high capacity (~59 MiB/s) with a marginal DA cost of ~\$0.010/MiB, close to Avail and Ethereum, but its short retention makes it the most expensive layer once cost is annualized. After Fusaka, Ethereum offers modest capacity at a cost comparable to Celestia and Avail, with the largest validator set and stake behind it.

The protocols were built for different jobs: Polkadot for parachain validation, NEAR for general smart-contract execution, Ethereum for censorship resistance and economic security, and Celestia and Avail as dedicated DA layers. Cost and throughput therefore do not rank them on their own; retention, trust assumptions, light-client verification and ecosystem integration matter as much.

7.3. Polkadot's Scaling Headroom

About 41 of Polkadot's 100 cores carried candidates on the sampled days, and the on-demand pool (~40 cores) had no inclusions on November 18, so the relay chain can absorb more parachain throughput without parameter changes. Estimated DA throughput (1.2–4.9 MiB/s, from parachain-side PoV sizes) is 1–3% of the ~167 MiB/s maximum. Bulk coretime prices would be expected to rise with demand; at the observed capacity price, cost is not a binding constraint. A more precise analysis would need compressed PoV sizes, which only active validators record.[1]

7.4. Implications for Rollup Developers

For optimistic rollups, the retention horizon constrains the choice of DA layer independently of cost. An optimistic rollup with a 7-day challenge period cannot rely on Polkadot (25 h) or NEAR (~1.5 d) without its own archive, and Celestia's 7-day sampling window leaves no margin. For the same reason, spot prices understate the cost of sustained availability on short-retention layers; the annualized cost (Section 4.2) counts the reposts needed to keep data available. In the window, minimum fees and token prices, more than congestion, set what rollups paid: Ethereum's post-Fusaka cost is consistent with the EIP-7918 reserve price, and Celestia's six-day partner load test at ~22% of capacity did not raise its median fee. Only Ethereum used a large share of its capacity, and only before the BPO forks; fee-market behaviour under sustained load remains to be observed.

7.5. Protocol Design Observations

Withholding enforcement. Slashing for omission faults requires attributable evidence, which none of the six designs produces; commission faults such as equivocation or invalid execution are attributable. Other enforcement models may be needed, for example insurance schemes, reputation systems with gradual stake reduction, or cryptographic mechanisms that make withholding unprofitable without having to attribute it.

Retention. Four protocols have a default retention horizon, and Avail and Espresso retain data unless operators enable pruning. Bounded retention is a deliberate choice that limits node storage growth; a DA layer is meant to provide a window for verification and fraud detection. Rollups and applications that need data beyond that window must store it elsewhere, for example on IPFS, Filecoin or Arweave.

[1] The `candidate_validation_pov_size` histogram exported by Polkadot validators.

7.6. Limitations and Threats to Validity

Demand-constrained measurements. Observed throughput reflects demand during one 92-day window; conclusions about maximum throughput must come from protocol parameters (Section 4.3).

Token prices. USD costs depend on token prices, which moved widely during the window (daily means: ETH $1,821–$3,357, TIA $0.31–$0.82), so another window would give different figures. Reporting VWAP and percentiles reduces this effect but does not remove it.

Polkadot PoV estimate. PoV sizes are not exposed on the relay chain, so we estimate them from 60 random blocks on each of 17 parachains (47% of inclusions) as `System.BlockWeight` proof size plus extrinsic bytes, weighted by each parachain's share of inclusions. This uncompressed proxy overstates the real PoV. Two parachains report a fixed proof size that looks like a weight reservation; with them the mean is ∼1.13 MiB per candidate (∼4.9 MiB/s), without them ∼0.29 MiB (∼1.2 MiB/s). Tables report the higher figure; both are uncompressed proof weights that include state witnesses, so they are not comparable to rollup payload bytes.

Espresso Mainnet 0. Mainnet 0 is a permissioned early deployment, so its cost, throughput and security figures cannot be compared on equal terms with permissionless, economically secured networks. We include it for architectural completeness.

Single observation window. A 92-day window is a snapshot, and behaviour under other market conditions, governance events or adoption levels may differ. The released tooling can be used to extend the window.

Cost measures. The six cost figures are of different kinds: Ethereum's is the blob base fee without the carrying transaction's execution gas; Celestia's quantiles use the gas price and exclude fixed PayForBlobs gas; NEAR's is a modelled marginal cost; Avail's is a per-transaction average; Polkadot's is a capacity price that also buys execution; Espresso's is a fixed placeholder. Spot fees also exclude batch overhead and compression, so they are not the effective cost to a rollup. The retention-adjusted cost (Table 7, Fig. 3(b)) depends on the retention definition in Section 4.2.

Data sources. Celestia data comes from the Celenium indexer, Ethereum data from the public BigQuery dataset, and prices from CoinGecko hourly data; we did not cross-check these against independent sources beyond spot checks of protocol parameters at in-window heights.

8. Conclusion

We measured six production DA protocols (Polkadot ELVES, Ethereum EIP-4844/PeerDAS, Celestia, Espresso Tiramisu, NEAR and Avail) with one methodology, using 92 days of mainnet data covering about 18.8 million blocks.

8.1. Summary of Findings

Window-mean utilization ranged from 0.18% (NEAR) to 35% (Ethereum). Ethereum ran at ∼60% of its maximum until the BPO forks raised capacity; the other five protocols stayed below 3%. Throughput comparisons across these networks therefore measure adoption rather than capacity.

Median DA cost spans nearly seven orders of magnitude, from $\sim\$3\times10^{-9}$/MiB (Espresso, whose 1 wei/byte fee is a Mainnet 0 placeholder) to $0.020/MiB (Celestia); the six measures differ in which fee components they include. On Ethereum, the blob fee left the 1-wei floor at Fusaka, coinciding with the EIP-7918 reserve price and before the BPO forks added capacity. Annualization reorders the ranking: NEAR's spot cost ($0.010/MiB) is close to Avail's and Ethereum's, but its annualized cost ($2.4/MiB/yr) is the highest because of its ∼1.5-day retention.

No protocol slashes for pure data withholding. Polkadot gates candidate inclusion on availability bitfields, Celestia uses bad-encoding fraud proofs, Ethereum uses peer scoring, Espresso relies on committee reputation, NEAR relies on its chunk-inclusion rule, and Avail slashes only for equivocation. Attributing withholding, an omission fault, to a specific actor is still unsolved.

No protocol guarantees permanent availability. Default retention ranges from 25 hours (Polkadot) to ∼18 days (Ethereum); Avail and Espresso retain data unless operators enable pruning; and among the four with a default horizon, only Ethereum's exceeds a 7-day optimistic-rollup challenge period with margin.

8.2. Future Work

Longitudinal extension: repeating these measurements as utilization rises would show supply-constrained behaviour that the current data cannot. *Stress testing*: controlled synthetic load would measure protocol limits directly, which observational data cannot do. *Withholding enforcement*: incentive-compatible enforcement for omission faults needs formal analysis. *Cross-layer analysis*: measuring end-to-end rollup performance (confirmation time, cost, finality) for each DA backend would show how the choice affects rollups. *PeerDAS at scale*: PeerDAS activated during our window; its effect on node bandwidth and on further BPO increases remains to be measured.

Data availability

Code and data are available at `https://github.com/Chainscore/score-da`, archived at `https://doi.org/10.5281/zenodo.23276229`.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used AI chat tools to assist with researching approaches to data collection. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] M. B. Saif, S. Migliorini, F. Spoto, A survey on data availability in layer 2 blockchain rollups: Open challenges and future improvements, Future Internet 16 (9) (2024) 315. doi:10.3390/fi16090315.

[2] V. Buterin, D. Feist, D. Loerakker, G. Kadianakis, M. Garnett, M. Taiwo, A. Dietrichs, EIP-4844: Shard blob transactions, Ethereum Improvement Proposals, no. 4844, created 2022-02-25; status Final (2022).
URL https://eips.ethereum.org/EIPS/eip-4844

[3] D. Ryan, D. Feist, F. D'Amato, H.-W. Wang, A. Stokes, EIP-7594: PeerDAS – peer data availability sampling, Ethereum Improvement Proposals, no. 7594, created 2024-01-12; status Final (2024).
URL https://eips.ethereum.org/EIPS/eip-7594

[4] M. Al-Bassam, LazyLedger: A distributed data availability ledger with client-side smart contracts, arXiv preprint arXiv:1905.09274 (2019). arXiv:1905.09274.
URL https://arxiv.org/abs/1905.09274

[5] J. Burdges, A. Cevallos, H. K. Alper, C.-D. Liu-Zhang, F. Shirazi, A. Stewart, R. Habermeier, R. Klotzner, A. Ordian, Efficient execution auditing for blockchains under Byzantine assumptions, Cryptology ePrint Archive, Paper 2024/961 (2024).
URL https://eprint.iacr.org/2024/961

[6] J. Bearer, B. Bünz, P. Camacho, B. Chen, E. Davidson, B. Fisch, B. Fish, G. Gutoski, F. Krell, C. Lin, D. Malkhi, K. Nayak, K. Shen, A. Xiong, N. Yospe, S. Long, The Espresso sequencing network: HotShot consensus, Tiramisu data-availability, and builder-exchange, Cryptology ePrint Archive, Paper 2024/1189 (2024).
URL https://eprint.iacr.org/2024/1189

[7] NEAR Protocol, The NEAR white paper, https://www.near.org/papers/the-official-near-white-paper, accessed: 2026-10-10 (2020).

[8] Avail Project, Avail: The data availability blockchain, Whitepaper, https://www.availproject.org/whitepaper, accessed: 2026-10-10 (2024).

[9] L. T. Thibault, T. Sarry, A. S. Hafid, Blockchain scaling using rollups: A comprehensive survey, IEEE Access 10 (2022) 93039–93054. doi:10.1109/ACCESS.2022.3200051.

[10] L. Heimbach, J. Milionis, The early days of the Ethereum blob fee market and lessons learnt, in: Financial Cryptography and Data Security (FC 2025), Part II, Vol. 15752 of Lecture Notes in Computer Science, Springer, 2026, pp. 53–71. doi:10.1007/978-3-032-07035-7_4.

[11] S. Lee, 180 days after EIP-4844: Will blob sharing solve dilemma for small rollups?, in: 2025 IEEE 45th International Conference on Distributed Computing Systems Workshops (ICDCSW), IEEE, 2025, pp. 487–492. doi:10.1109/ICDCSW63273.2025.00096.

[12] M. Al-Bassam, A. Sonnino, V. Buterin, I. Khoffi, Fraud and data availability proofs: Detecting invalid blocks in light clients, in: Financial Cryptography and Data Security (FC 2021), Part II, Vol. 12675 of Lecture Notes in Computer Science, Springer, 2021, pp. 279–298, preprint: arXiv:1809.09044. doi:10.1007/978-3-662-64331-0_15.

[13] M. Hall-Andersen, M. Simkin, B. Wagner, Foundations of data availability sampling, IACR Communications in Cryptology 1 (4) (2025). doi:10.62056/a09qudhdj.

[14] I. S. Reed, G. Solomon, Polynomial codes over certain finite fields, Journal of the Society for Industrial and Applied Mathematics 8 (2) (1960) 300–304. doi:10.1137/0108018.

[15] A. Kate, G. M. Zaverucha, I. Goldberg, Constant-size commitments to polynomials and their applications, in: Advances in Cryptology – ASIACRYPT 2010, Vol. 6477 of Lecture Notes in Computer Science, Springer, 2010, pp. 177–194. doi:10.1007/978-3-642-17373-8_11.

[16] C. Cachin, S. Tessaro, Asynchronous verifiable information dispersal, in: Proceedings of the 24th IEEE Symposium on Reliable Distributed Systems (SRDS'05), IEEE, 2005, pp. 191–202. doi:10.1109/RELDIS.2005.9.

[17] J. Burdges, A. Cevallos, P. Czaban, R. Habermeier, S. Hosseini, F. Lama, H. K. Alper, X. Luo, F. Shirazi, A. Stewart, G. Wood, Overview of Polkadot and its design considerations, arXiv preprint arXiv:2005.13456 (2020). arXiv:2005.13456.
URL https://arxiv.org/abs/2005.13456

[18] P. Jayanthi, T. Wahrstätter, S. Calder-Mason, A. Davis, A. Dietrichs, EIP-7691: Blob throughput increase, Ethereum Improvement Proposals, no. 7691, created 2024-04-17; status Final (2024).
URL https://eips.ethereum.org/EIPS/eip-7691

[19] A. Elowsson, B. Adams, F. D'Amato, EIP-7918: Blob base fee bounded by execution cost, Ethereum Improvement Proposals, no. 7918, created 2025-03-25; status Final (2025).
URL https://eips.ethereum.org/EIPS/eip-7918

[20] K. Croman, C. Decker, I. Eyal, A. E. Gencer, A. Juels, A. Kosba, A. Miller, P. Saxena, E. Shi, E. G. Sirer, D. Song, R. Wattenhofer, On scaling decentralized blockchains, in: Financial Cryptography and Data Security: FC 2016 International Workshops, BITCOIN, VOTING, and WAHC, Vol. 9604 of Lecture Notes in Computer Science, Springer, 2016, pp. 106–125. doi:10.1007/978-3-662-53357-4_8.

[21] M. Hall-Andersen, M. Simkin, B. Wagner, FRIDA: Data availability sampling from FRI, in: Advances in Cryptology – CRYPTO 2024, Part VI, Vol. 14925 of Lecture Notes in Computer Science, Springer, 2024, pp. 289–324. doi:10.1007/978-3-031-68391-6_9.

[22] M. Yu, S. Sahraei, S. Li, S. Avestimehr, S. Kannan, P. Viswanath, Coded Merkle tree: Solving data availability attacks in blockchains, in: Financial Cryptography and Data Security (FC 2020), Vol. 12059 of Lecture Notes in Computer Science, Springer, 2020, pp. 114–134. doi:10.1007/978-3-030-51280-4_8.

[23] P. Sheng, B. Xue, S. Kannan, P. Viswanath, ACeD: Scalable data availability oracle, in: Financial Cryptography and Data Security (FC 2021), Part II, Vol. 12675 of Lecture Notes in Computer Science, Springer, 2021, pp. 299–318. doi:10.1007/978-3-662-64331-0_16.

[24] K. Nazirkhanova, J. Neu, D. Tse, Information dispersal with provable retrievability for rollups, in: Proceedings of the 4th ACM Conference on Advances in Financial Technologies (AFT '22), ACM, 2022, pp. 180–197. doi:10.1145/3558535.3559778.

[25] B. Fisch, A. Lazzaretti, Z. Liu, L. Yang, Permissionless verifiable information dispersal (data availability for Bitcoin rollups), in: 2025 IEEE Symposium on Security and Privacy (SP), IEEE, 2025, pp. 1983–2001. doi:10.1109/SP61157.2025.00248.

[26] ethereum.org, Danksharding, https://ethereum.org/roadmap/danksharding/, accessed: 2026-10-10 (2024).

[27] A. Chaudhuri, S. Basak, C. Kiraly, D. Ryajov, L. Bautista-Gomez, On the design of Ethereum's data availability sampling: A comprehensive simulation study, in: Proceedings of the 6th Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS), IEEE, 2024, pp. 1–4, extended version: arXiv:2407.18085. doi:10.1109/BRAINS63024.2024.10732486.

[28] M. Pigaglio, O. Ascigil, M. Król, F. Lange, K. Peeroo, S. Rene, R. Sadre, V. Stankovic, E. Rivière, PANDAS: Peer-to-peer, adaptive networking allowing data availability sampling within Ethereum consensus timebounds, in: Proceedings of the 26th International Middleware Conference (Middleware '25), ACM, 2025, pp. 167–179. doi:10.1145/3721462.3770769.

[29] D. Crapis, E. W. Felten, A. Mamageishvili, EIP-4844 economics and rollup strategies, in: Financial Cryptography and Data Security. FC 2024 International Workshops, Vol. 14746 of Lecture Notes in Computer Science, Springer, 2024, pp. 135–149. doi:10.1007/978-3-031-69231-4_10.

[30] S. Park, B. Mun, S. Lee, W. Jeong, J. Lee, H. Eom, H. Jang, Impact of EIP-4844 on Ethereum: Consensus security, Ethereum usage, rollup transaction dynamics, and blob gas fee markets, IEEE Transactions on Network Science and Engineering 13 (2026) 1294–1310. doi:10.1109/TNSE.2025.3594281.

[31] Polkadot Developer Docs, Overview of the Polkadot relay chain,

https://docs.polkadot.com/reference/polkadot-hub/consensus-and-security/relay-chain/, accessed: 2026-10-10 (2025).